

Using Transparent Neural Networks and Wearable Inertial Sensors to Generate Physiologically-Relevant Insights for Gait

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Abstract—Neural networks have been successfully applied to a wide range of human motion analysis topics in combination with wearable sensor data. However, their computation process is not readily comprehensible. Alternatively, many of the model interpretation efforts do not provide physiologically-relevant insights, thus still limiting their use in clinical settings. In this work, we take gait modifications under fatigue and cognitive task performance as a use case to present how in-depth investigations of neural networks can be performed using wearable sensor data. We collected walking data from 16 young healthy individuals in unfatigued and fatigued states and under single- (walking only) and dual-task (walking while concurrently performing a cognitive task) conditions using inertial measurement units. Convolutional neural networks were able to identify both fatigue and dual-task gait patterns with high classification accuracy. To interpret the model, the importance of each time step in the input time series was visualized using Layer-wise Relevance Propagation. The visualization revealed highly individualized gait changes among participants, as well as changes at precise time steps of the input signal that allow further investigations to infer potential underlying mechanisms. Our methods enable in-depth analysis of human movement using transparent neural networks with data collected from unobtrusive, mobile wearable sensors.

Index Terms—Artificial Neural Networks, Explainable AI, Gait Analysis, Inertial Measurement Units, Data Visualization

I. INTRODUCTION

Neural networks are powerful tools for studying human movements, such as human activity recognition, human motion prediction, or early disease detection. In many cases, the computation process of neural networks is not readily comprehensible, or the models are not interpreted in a way that yields physiologically-relevant insights. Gait analysis is a prominent topic for human movement analysis using machine learning, and mobile wearable sensors enable unobtrusive recordings of gait patterns in daily life settings.

Muscle fatigue and multi-tasking are two of the most relevant factors involved in daily life walking, irrespective of the population under investigation. Muscle fatigue leads to reduced balance and may increase the risk of falls and/or injuries [1] [2]. Performing a cognitive task concurrently while

walking (dual-task, such as walking while talking on a mobile phone) simulates walking in daily life environments [3]. Therefore, understanding the mechanisms of gait modifications under these conditions will support the reduction of injuries or fall risk in real-life situations.

Inertial measurement units (IMUs) are low-cost, mobile sensors commonly used for human gait analysis. Gait characteristics represented by joint angles [4] or spatiotemporal gait parameters [5] (e.g., walking speed, stride length, or stride time) can be calculated from the raw IMU data. These methods require strong domain knowledge, and sometimes extensive, manual data preprocessing is necessary to obtain accurate gait parameters. Alternatively, the raw acceleration and angular velocity data recorded with IMUs can be directly used by deep learning algorithms for gait parameter estimation [6] [7], gait phase recognition [8], or gait classification [9] with high prediction accuracy. For example, long-short-term memory (LSTM) and convolutional neural networks (CNN) have been used to identify fall-risk [10] [11]. Despite the high performance of those algorithms, the explainability of the models is often unexplored. Most of the studies classifying fatigue and non-fatigued gait patterns used the support vector machine (SVM) classifier. Karvekar et al. achieved 91% two-class classification accuracy using significant features selected according to the t-test, however, the contribution of each feature to the classification outcome was not explored [12]. Similarly, Zhang et al. used an SVM model with a set of IMU-derived significant features based on t-test, and achieved an accuracy of 0.93 using these features [13]. In their work, the authors visualized the decision boundaries of the classifiers, which provided additional information about the models, but was not sufficient for understanding what aspects of the gait has changed under fatigue condition. There are algorithms that compute the quantitative or ranked importance of input features for various models, such as feature coefficients for linear models, or permutation feature importance for general machine learning models. However, with these classical feature im-

portance measures, it remains challenging to identify exactly which part of the time series is important for classification, thus limiting further investigation into the movement that is affected.

In the context of Explainable AI, methods to enable the interpretation of neural networks (i.e., transparent neural networks) have proven their practicability for gait classification. Dindorf et al. used Local Interpretable Model-agnostic Explanations (LIME) to identify important features differentiating the gait of hip arthroplasty patients from healthy controls. However, as discussed before, model interpretation with discrete features or input instances does not pinpoint the exact movement changes which could be clinically relevant. Horst et al. [14] and Slijepcevic et al. [15] interpreted neural network classifiers using Layer-wise Relevance Propagation (LRP), in which output logits backpropagate to the input layer, and are decomposed to each input feature and each sample point into relevance scores. As a result, the LRP relevance score of an input sample point is proportional to its impact on the classification decision [17]. In terms of time series inputs such as raw IMU signals, it is, therefore, possible to identify the contribution of each input time step to the classification outcome using the LRP scores assigned to this time step. Consequently, the exact movement that distinguishes between classes can be identified. Indeed, in studies by Horst et al. [14] and Slijepcevic et al. [15], time series of joint angle and/or ground reaction force were nicely visualized with LRP relevance scores, so that the important segments distinguishing different classes were identified and confirmed by clinical experts. Nonetheless, the data collection setup in laboratory settings with force plates or full-body optical markers limits their applications in real life. Table I summarizes related work classifying gait patterns using raw IMU signals, or interpreting gait data using LRP or related methods.

To the best of our knowledge, there have been no reports of using transparent neural networks to investigate the gait kinematics under fatigue and dual-task conditions using IMU data. In this study, we demonstrate the feasibility of associating mobile sensor signals to the individualized kinematics, and then use this information to infer possible underlying physiological/neuromuscular mechanisms.

II. MATERIALS AND METHODS

A. Data Acquisition

Sixteen young healthy participants (eight females, eight males) were recruited to participate in this study. Study eligibility was assessed with the Physical Activity Readiness Questionnaire (PAR-Q), participants who replied "yes" to any of the questions—indicating medical restrictions to perform physical exercises—were excluded from this study. Participant characteristics are summarized in Table II.

The data collection procedure consisted of two visits (referred to as visits A and B in the following descriptions) which were seven days apart. During visit A, the participants performed two 6-minute walking sessions with a one-way

distance of 35 m, separated by a fatigue protocol. After the first walking session, the participants performed a sit-to-stand all-out fatigue protocol, in which they wore a weighted vest matched to 30% of their individual body mass, and repeatedly stood up from a chair at a preferred rapid pace until failure. The fatigue status was confirmed using blood lactate concentration measurement with blood sampled from the earlobe [18], and Borg Rating of Perceived Exertion [19]. Immediately following the fatigue protocol, the participants repeated the 6-minute walking session. The procedures during visit B were the same as those during visit A, except that the participants performed a secondary cognitive task concurrently during both walking sessions (i.e., the dual-task condition): the participants were asked to serially subtract seven from a four-digit number provided by the experimenter (randomly sampled between 3000 and 9000). The sequence of visits A and B were randomized for all participants. In total, four walking sessions were performed by each participant: single-task control (ST-Control), single-task fatigue (ST-Fatigue), dual-task control (DT-Control), and dual-task fatigue (DT-Fatigue).

The complete dataset was collected using nine Physilog®5 IMUs (Gait Up, Switzerland) attached to the feet (LF and RF), legs (LL and RL), wrists (LW and RW), the sacrum (SA), the sternum (ST) and the head (HE) of the participants. For the purpose of this study, we only consider three sensors: LF, RF, and SA. Three-axis acceleration (range: $\pm 16\text{ g}$) and angular velocity (range: $\pm 1000\text{ degrees/sec}$) were recorded at a sampling rate of 128 Hz. All IMU signals were synchronized using the synchronization option provided by the manufacturer. An audio recorder was attached close to the left collar bone to record answers from the cognitive task. The experimental setup is illustrated in Figure 1.

The study was approved by the ethics committee of the University of Potsdam (63/2020), and all experiments were conducted according to the latest revision of the declaration of Helsinki. All participants provided written consent prior to data collection.

B. Classification with CNN

1) *Data Preprocessing:* We experimented with different combinations of sensor placements (LF, SA, LF + SA, or LF + RF + SA), and focus on presenting the results from the combination of all three sensor placements (LF + RF + SA). The input data for each individual consists of time series of triaxial acceleration and angular velocity signals from IMUs located at LF, RF, and SA, thus resulting in 18 features (i.e., $3 \times 2 \times 3$). The raw IMU signals were segmented into strides by initial contact events detected from the angular velocity of the LF sensor using the method proposed by Tunca et al. [5], and padded to the same length. Non-overlapping windows of single strides were fed into the CNN so that in later steps for model interpretation, the results can be interpreted with respect to a gait cycle. A third-order Butterworth low-pass filter with a cutoff frequency at 10 Hz was used to smooth the input time series data to reduce sensor noise. To compensate for the differences in numerical value ranges between the accelera-

TABLE I: Related work on gait classification using IMU data or transparent neural networks.

Ref.	Classifica- tion Target	IMU Placement	Subjects	Features	Explain- ability	Classifier	Result
Zhang et al. [13]	Fatigue	Ankle, chest	17 healthy subjects	Descriptive features from raw accelerometer	-	SVM	Acc = 0.97
Karvekar et al. [12]	Fatigue	Shank	24 healthy subjects	Descriptive features from raw accelerometer, participant characteristics	-	SVM	Acc = 0.91
Tunca et al. [10]	Fall risk	Feet	76 neurological disorder patients	Raw inertial data, Gait parameters	-	LSTM	Acc = 0.76 ¹
Aicha et al. [11]	Fall risk	Lower back	296 elderly adults	Raw accelerometer, Gyroscope signals	-	LSTM+CNN	AUC = 0.75
Horst et al. [14]	Identity	-	57 healthy subjects	Body joint angles, ground reaction force	LRP	CNN	Acc = 1.0
Sljepcevic et al. [15]	Gait disorders	-	132 lower-body gait disorder patients 62 controls	Ground reaction force	LRP	CNN, SVM, MLP	Acc $\approx$ 0.9
Dindorf et al. [16]	Gait after hip surgery	Lower extremities	20 hip arthroplasty patients, 27 controls	Raw inertial data, descriptive statistics	LIME	MLP, SVM, RF	Acc = 1.0

¹ Accuracy using raw inertial data = 0.76, accuracy using gait parameters = 0.92.

TABLE II: Participant characteristics.

Feature	Mean $\pm$ SD	Min	Max
Age (years)	27.1 $\pm$ 3.8	21	35
Body Mass (kg)	71.2 $\pm$ 12.2	54	103
Height (cm)	173.8 $\pm$ 8.6	158	190
Leg Length (cm)	83.9 $\pm$ 3.8	78	94



Fig. 1: Experimental setup. Left: sensor setup; right: sit-to-stand fatigue protocol.

tion (g) and angular velocity (degrees/sec), the raw signal data were normalized with the commonly used *z* transformation. The normalization parameters (i.e., μ and σ) were calculated only from the training data set, and then applied to all sets to avoid introducing information about the validation and test sets into training data.

2) *CNN Architecture*: A one-dimensional convolutional neural network (1D-CNN) was implemented for classifying

between ST-Control and ST-Fatigue conditions, as well as between ST-Control and DT-Control conditions. The structure of the 1D-CNN is visualized in Figure 2. The *1D-CNN layer* has 32 filters and uses the ReLu activation function. The filters have a window size of 5 for each of the 18 features (i.e., $32 \times 5 \times 18$). The convolution stride length is 1. Subsequently, a *GlobalMaxPooling layer* is applied that shrinks the feature map to 32×1 . Finally, a *dense layer* predicts the two classes with softmax activation. The initial learning rate is 0.02 with a *learning rate scheduler* that decreases the learning rate by the factor of 0.5 until a minimum learning rate of 0.001 is reached or the validation loss has not improved for two subsequent epochs. Further, *early stopping* is applied such that the training stops if the validation loss has not improved for three consecutive epochs. Our preliminary experiments comparing different neural network architectures indicated that the above-described model is sufficient for the task, as more complex models did not improve the classification results.

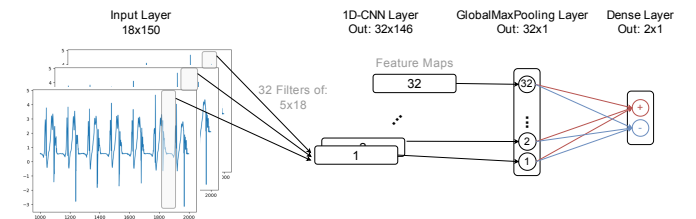


Fig. 2: Structure of the 1D-CNN architecture. The input is one gait cycle with 18 features. The CNN layer has 32 filters with a window length of 5 with the ReLu activation function, followed by a GlobalMaxPooling layer and a dense layer to predict the two classes.

3) *Model Training and Evaluation*: Personal-specific models were trained and evaluated in this study. It is known that gait characteristics are highly individual. For example, Horst et al. demonstrated in their research that an SVM

classifier was able to identify the gait characteristics of 128 individuals with an accuracy of over 0.99, and subsequent one-year follow-up classification also achieved the same accuracy [20]. Moreover, there is evidence that gait responses to interventions are highly individual, both for sports and medicine [21] [22]. Related work on gait classification using a deep Long-Short-Term-Memory (LSTM) neural network and raw IMU data reported accuracy of 0.76 [10]. Consistent with this knowledge, in our preliminary experiments, generalized models with more complex architecture and leave-one-person-out cross-validation resulted in the best average accuracy of only 0.77. A large inter-personal variation could be observed in the classification results. Further interpretation of the model with the aim of inferring clinically-relevant mechanisms is only meaningful when the classification results are reliable (with sufficiently high accuracy). Therefore, we focus on analyzing gait changes from each individual participant by training and evaluating personalized models using the data of an individual. For each personalized model, 10% of the data was left out for testing. Five-fold cross-validation was performed on the remaining 90% of the data. Thus the train, validation, and test sets contain 72%, 18%, and 10% of the data, respectively.

C. Visualization using Layer-wise Relevance Propagation

We used Layer-wise Relevance Propagation (LRP) for neural network interpretation, with the *iNNvestigate* library developed by Alber et al. [23]. With the LRP method, one relevance score is calculated for each time step within each input time series. This relevance score is the amount of activation that the single time step contributes to the logits of the positive class neuron in the output layer of the neural network. Due to the small number of layers in our model architecture, the LRP-epsilon rule [24] was used for all layers. Formal description of the association of a relevance score $R_i^{(l)}$ (neuron i in layer l) with relevance scores in its downstream layer $l+1$ is given by:

$$R_i^{(l)} = \sum_j \frac{z_{ij}}{\sum_{i'} z_{i'j} + \epsilon \text{sign}(z_j)} R_j^{(l+1)} \quad (1)$$

with

$$z_{ij} = a_i^{(l)} w_{ij}^{(l,l+1)}, \quad (2)$$

where i' denotes neurons in the same layer, $a_i^{(l)}$ is the activation of neuron i in layer l and $w_{ij}^{(l,l+1)}$ is the weight parameter connecting neuron i in layer l with neuron j in layer $l+1$. Thus, the normalized relevance of each neuron j in the layer $l+1$ (i.e., $R_j^{(l+1)}$) is propagated to each neuron i in the preceding layer l (that is connected with a weight to it). This propagated relevance $R_j^{(l+1)}$ is normalized by the proportion of its quantity z_{ij} relative to the sum of all z of all neurons at the layer l that are connected to neuron j in layer $l+1$. Thus, a neuron that is responsible for most activation for a given downstream neuron will receive the most relevance from that downstream layer neuron. The term $\epsilon \in \mathbb{R}_{>0}$ is introduced to increase numerical stability, and the sign of ϵ depends

on the sign of the output activation. Unseen data (i.e., the test set) and trained models were used to generate relevance scores. Based on the obtained relevance scores, we developed two visualization approaches to interpreting the classification models based on LRP: feature importance bar plots and time series plots.

To determine the overall importance of a feature, the LRP scores of all time steps within one input time series example from that feature were summed up, and the sums for all input examples were averaged. This average value represents feature importance for the classification decision, and was visualized in a bar plot for comparison. Features whose importance is above the average of all positively-contributing features were identified as top important features, and their input time series were analyzed further using the LRP time series plot.

To examine in more detail which parts of the movement profile were important for the neural networks, the LRP time series plot visualizing the raw sensor signal of one feature (e.g., acceleration y-axis), inspired by [14], was created. Each time step in the time series was colored according to its LRP score. Colored time series of all strides within one walking session from the same participant were then aggregated into one plot, where recurring patterns in the LRP scores are enhanced. To incorporate domain knowledge, timestamps of key gait events (i.e., initial contact and foot off) were indicated on the same plot. With this information, the patterns in the LRP scores can be associated with movement of the foot, enabling further interpretations of the underlying causes for the change of movement.

III. RESULTS AND DISCUSSION

A. Classification Performance

Classification performance for the personalized models is summarized in Table III. For the great majority of the participants, the accuracy for classifying fatigue was above 0.95. For some participants, the accuracy was even as high as 1.0, indicating that the models were able to identify clear distinctions between gait characteristics in unfatigued and fatigued states. The average test accuracy for the prediction of dual-task was 0.999 ± 0.01 . The combination of multiple IMU data sources (e.g., LF and RF, or LF, RF, and SA) slightly improved the classification performance: depending on the IMU placement, data from one single IMU was quite sufficient for the models to reach a high average accuracy ranging from 0.94 to 1.0. Nevertheless, there are noticeable variations between the performances of the personalized models.

B. Transparent Neural Network Visualization

Overview of feature importance was illustrated with bar plots of aggregated LRP scores. Figure 3 shows feature importance from classification models of three example participants. It can be clearly observed that despite the high classification accuracy of all models, distributions of important features differ among these individuals. Similarly, feature importance from all other participants' models is also highly individualized.

TABLE III: Test accuracy of personalized CNN models for classifying fatigue.

Data Source	sub01	sub02	sub03	sub05	sub06	sub07	sub08	sub09	sub10	sub11	sub12	sub13	sub14	sub15	sub17	sub18	Mean $\pm$ SD
LF	0.99	0.89	0.97	0.97	0.94	0.95	0.90	1.00	0.92	0.99	0.99	0.95	0.92	0.98	0.71	0.98	0.94 $\pm$ 0.07
SA	0.99	1.00	1.00	0.99	1.00	1.00	1.00	1.00	1.00	0.99	1.00	1.00	1.00	0.99	1.00	1.00	1.00 $\pm$ 0.01
LF+RF	1.00	0.96	0.97	0.98	0.90	0.95	0.90	1.00	0.98	1.00	1.00	0.94	1.00	0.97	0.89	1.00	0.96 $\pm$ 0.04
LF+RF+SA	0.99	0.99	0.99	1.00	1.00	1.00	0.98	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00 $\pm$ 0.01

LF: left foot, RF: right foot, SA: sacrum

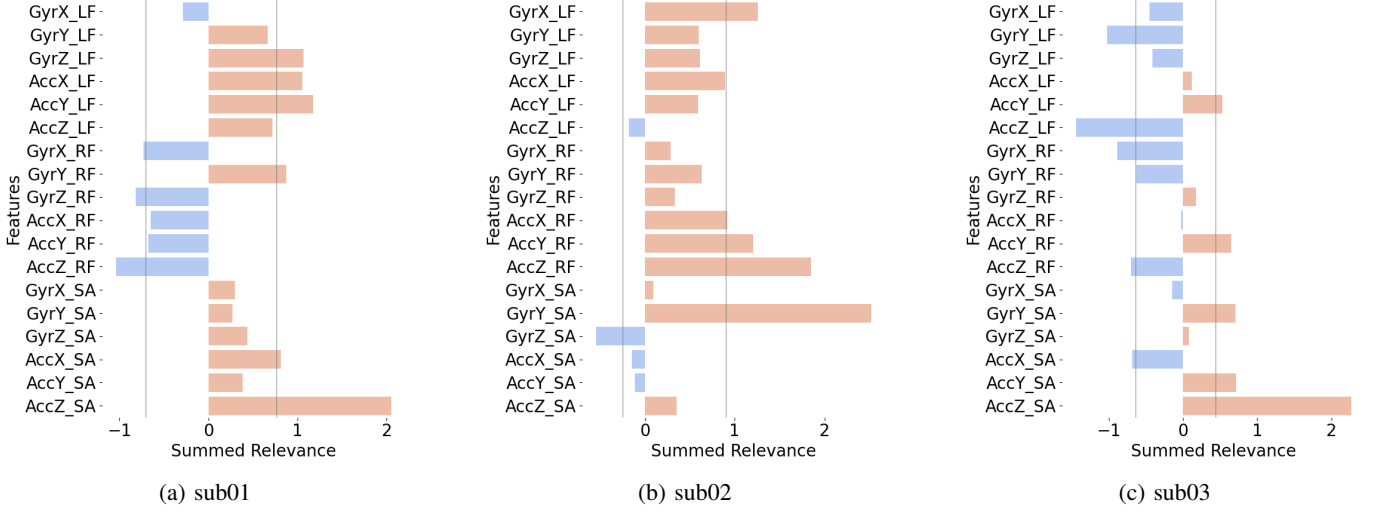


Fig. 3: Barplots of aggregated Layer-wise Relevance Propagation (LRP) scores from three examples of the personalized models. Each bar visualizes the overall feature importance of one feature for the fatigue class test samples. Blue bars are LRP sums of negatively contributing features, orange bars are positively contributing features. The vertical line indicates the average LRP sum of positive and negative scores, respectively. The important features (i.e., supporting fatigue class above the average value) for the models are different among the participants.

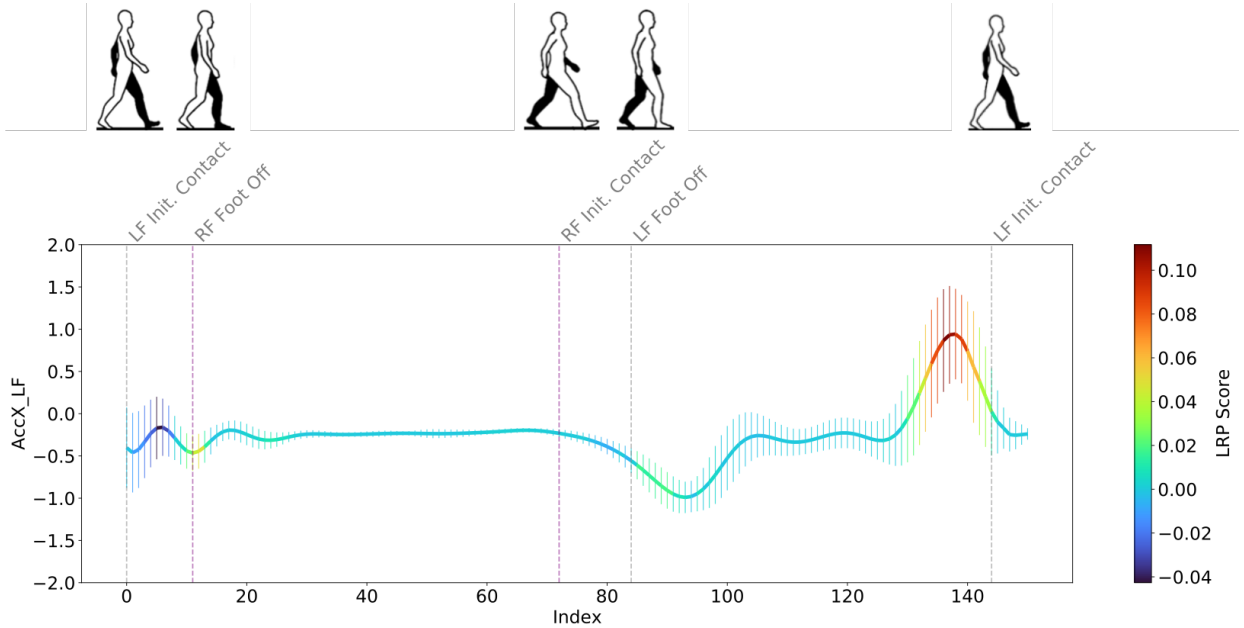


Fig. 4: Aggregated LRP time series plots of the mediolateral feet acceleration (AccX LF) for one person (sub01). The acceleration signal over the entire walking session was segmented by initial contact, the time series line represents mean values of each time step within the stride, and error bars represent standard deviation. Averaged gait events from all strides are labeled with vertical dotted lines.

To interpret the neural network models for gait classification in-depth, we created LRP plots for each personalized model. As an example, the LRP plot for one model (sub01) and fatigue classification (using data from ST-Fatigue condition) is illustrated in Figure 4. Averaged gait events from all strides were labeled with vertical lines. It was observed that the mediolateral acceleration (AccX) peak immediately before initial contact, namely, the terminal swing phase, contributed most to the correct classification result. When regarding important features from all individualized models, the effects of fatigue and dual-task are different. Eight participants' models share AccX LF/RF as one of their most important features, and seven of the stride time series exhibit this pattern at the terminal swing phase (86% of the models). In contrast, for the ten personalized dual-task classification models that share AccX LF/RF as important features, only five exhibited the terminal swing phase as an important segment for classification (50% of the models). Whereas for the other five models, no common patterns could be observed.

C. Interpretation of Transparent Neural Networks

We observed heterogeneous feature importance for personalized classification models, which indicates that changes in gait characteristics are highly individualized. Interestingly, the most important features from LRP analysis for dual-task classification are identical to those used for fatigue classification, namely, mediolateral and anterior-posterior acceleration of the foot, as well as the mediolateral acceleration of the sacrum. Detailed examination of the time series LRP plots revealed that adding a cognitive task resulted in greater between-subject variability of the observed kinematic changes than inducing muscle fatigue. Accordingly, it is reasonable to assume that fatigue- and dual-task-related changes might have different underlying mechanisms or that the same mechanisms contribute differently to the observed changes. Here, we provide one example of exploring the possible mechanisms based on the analysis of the LRP time series plots. We observed that mediolateral acceleration during the terminal swing phase is important for classification for several participants, as illustrated in Figure 4. From a biomechanical perspective, one of the main contributors to foot movement during the swing phase is the Tibialis Anterior (TA) muscle [25]. During stance, this muscle also helps to maintain the center of pressure of the heel along the mediolateral axis [26]. When the participants execute the fatigue protocol, the TA muscle helps to align the body's center of mass over the base of support (i.e., feet), thus maintaining balance [27]. Subsequently, a fatigued TA muscle may result in decreased stability during walking, which could have contributed to gait pattern changes in the mediolateral direction [26]. In contrast, the increased attentional demands in the dual-task conditions may have led to participants using compensation strategies to increase stability while walking, for example, expanding the base of support during walking by toeing out, which again affords TA activation [28]. From the LRP time series plots, we could infer that about half of the participants whose terminal swing phase contributed to the

classification result might have utilized this strategy, while the others might have employed different means of compensation.

D. Limitations and Future Work

In this work, we demonstrated how data collected using a small wearable sensor setup can generate physiologically-relevant insights using transparent neural networks. As a proof of concept, in this paper, we only focused on presenting one specific visualization and analysis process, namely, the in-depth investigation of the effects of fatigue on gait characteristics of one individual. Similar methods can be applied to different use cases requiring more complex neural network architectures.

Our model interpretations are supported by multiple studies on the anatomy and function of the human gait. Nonetheless, it is worth emphasizing that the aim of this study is to demonstrate the model interpretation method with gait as a use case. We cannot make assertions about the underlying physiological mechanisms, since experimental evidence from the same participants is missing, and other competing theories to explain the potential underlying mechanisms may exist. As a future validation study, the interpretations demonstrated in this work can be confirmed using lab-based measurements such as electromyography (EMG) [29].

Given the amount and variety of our visualization results (fatigue and dual-task classification, six features) from all individual models, further explorations may yield additional interesting findings. In addition, since walking involves the entire body, further analyses using data from additional sensor placements could provide a more comprehensive overview of changes in gait and balance, as well as potential different compensation strategies employed by different individuals. This could be the topic of another study that is focused on investigating mechanisms of fatigue and dual-task on gait.

IV. CONCLUSION

In this study, we demonstrate that with a small wearable sensor setup, it is possible to perform an in-depth investigation of the effects of fatigue and cognitive task performance by leveraging transparent neural networks. Our analyses range from classification outcomes to feature importance, through pin-pointed movement profiles within the time series, and finally to inferring possible anatomical and functional correlates. Our data collection setup with mobile, unobtrusive wearable sensors enables convenient data collection in free-living environments. With gait as a use case, our methods can be extended to various human movement data collected from daily life and provide insights into potential underlying mechanisms.

ACKNOWLEDGMENT

The authors would like to thank all participants in this study. We would also like to thank Justin Albert and Torsten Rackoll for their help with drafting the ethics proposal, Arnd Gebel, Kristina Kirsten, Sarah Wernicke and Jonas Chromik for helping to develop the fatigue protocol, and Pascal Hecker for proofreading.

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